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A series of horizontal lines for writing answers, spanning the width of the page.

## A] DOS Attack.

Denial of Service attack.

A DOS attack is when you overwhelm a server by sending it more pings than it can process so it stops functioning temporarily. This could be prevented by rate-limiting. With DOS attack the service would temporarily be downed until all the pings are cleared or the server is reset.

## B] Active and passive attack

### → Active attack

- It is an attack which may cause harm, the victims will acknowledge that failure is happening.
- The attacker can actively change the data in the server etc.

### → Passive attack

- It is an attack that is passive. The victims might not know about it.
- The attacker is in the system but doesn't change anything, note or eavesdropping.



## C) CVE.

Common Vulnerability Exploit.

It is a record id for a vulnerability or exploit found for any major services. It is a public record for all the vulnerabilities, exploits found.

When recorded it will be named like CVE-22708.

Any one can look up for any CVE to cross check if those systems have any vulnerabilities.

## D) Security Association (SA)

Security Association is the Association of Security.

They ensure data security, Confidentiality and integrity.

It is made sure that ~~Privacy~~ data is encrypted and ~~Secure~~ Secure by the association.

## E) S/MIME

Secure / Multipurpose Mail encryption.

- It is a protocol that can be followed in mail.
- Its purpose is to ensure no data loss by encryption.
- It secures the connection from server to client.
- It ensures no one can access ~~the data~~ encrypted data.
- It is only accessed by sender and receiver.

## F) PGP

Pretty Good ~~Protocol~~ Privacy

It is a protocol used in mail for data integrity and privacy.

PGP operations

- 1) Encryption
- 2) Providing Confidentiality
- 3) Digital Certificate
- 4) Verifying the structure.

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## A Popular threats over confidentiality

There are many threats so that is why we have

CIA Triad

- Confidentiality
- Integrity
- Availability

A few threats over Confidentiality are like.

- Identity theft
- Company secrets theft
- Data theft.

The consequence for such actions is a tremendous loss for the user or company.

If important confidential data is leaked then it falls in the wrong hands then it is very bad.

for that we can implement countermeasures like

- Encryption
- Hashing
- Cryptograph
- Physical Security

With the implementation of countermeasures, the security of the confidential data is improved.

## B SSL.

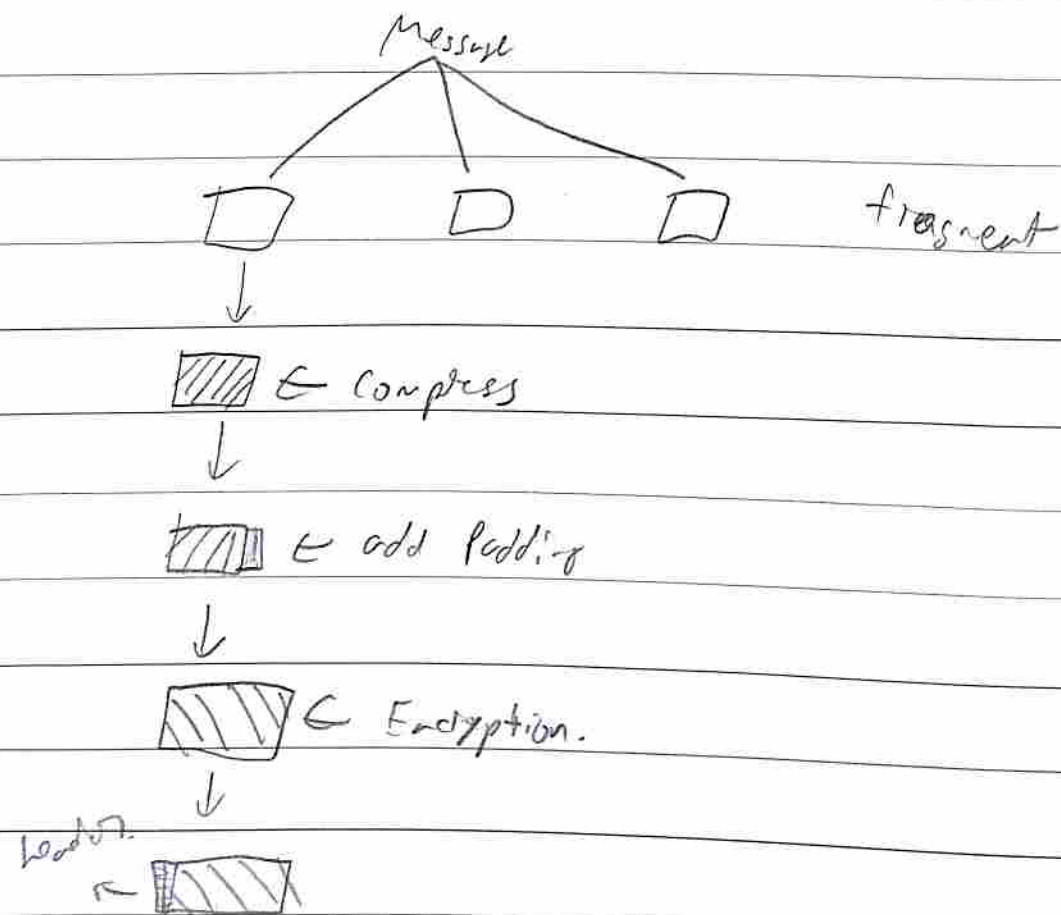
Secure Socket Layer.

SSL is mainly used at the transport layer. It is used for signing and encryption.

The record protocol operation in Secure Socket Layer is used to record all of the activity happening in the SSL layer.

It is important to record the activity in the layer.





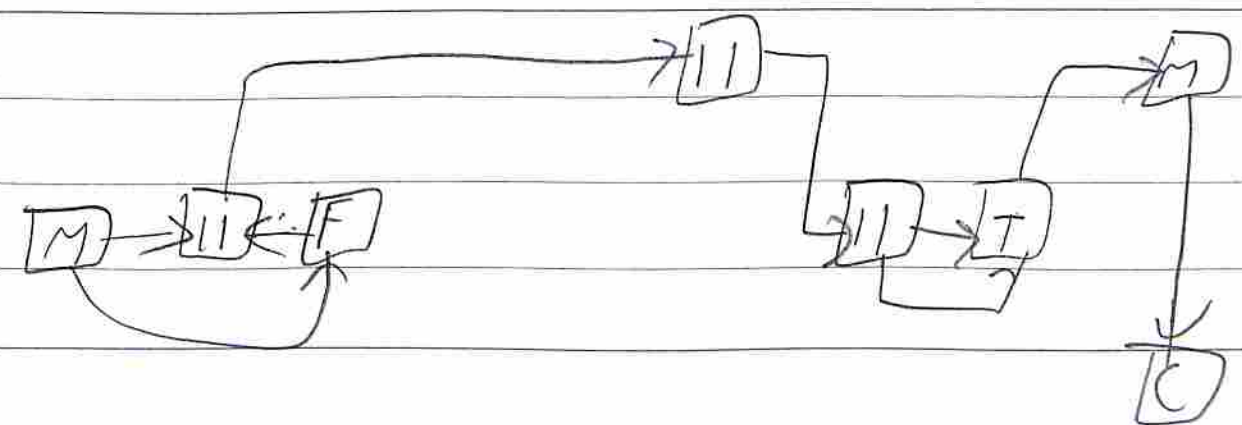
The Record protocol is important in SSL as it will record all activity happen in SSL, with that we can analyze the traffic and implement appropriate measures. Also the SSL. This will help in case of intruder.

## 7. Public and Private key usage in PGP

• TO provide the utmost ~~and~~ secure and possibly confidentiality, PGP follows asymmetric encryption to encrypt

• TO encrypt. PGP will share its public key so that the sender will encrypt it with its public key

• to decrypt PGP will receive the cipher data and decrypts it with its own private key to access data.



This is the general structure of the public and private key of PGP

D/

DKIM

Domain Key Index Message

• DKIM is implemented at DNS level.

• It is used to secure the mails so that there is no data loss

• It encrypts the data so that data integrity is maintained

• It ensures only the authorized receiver can access it and not ~~intruders~~ intruders.

• It encrypts the data so that ~~sender~~ receiver can understand the ~~sender's~~ encrypted data

A/ Port Scanning NMAP

When we execute port scanning we check for all of the open ports in the network or the server.

This is a healthy practice because when we execute port scanning we can see all of the open ports accessible to our server and we can identify any vulnerability.

In the other case when performed on a target the same open ports of their server is visible. And if they hadn't configured it properly we can access their data through the open ports.

A common tool used for this is NMAP, short for network mapper. It is easy to use and understand.



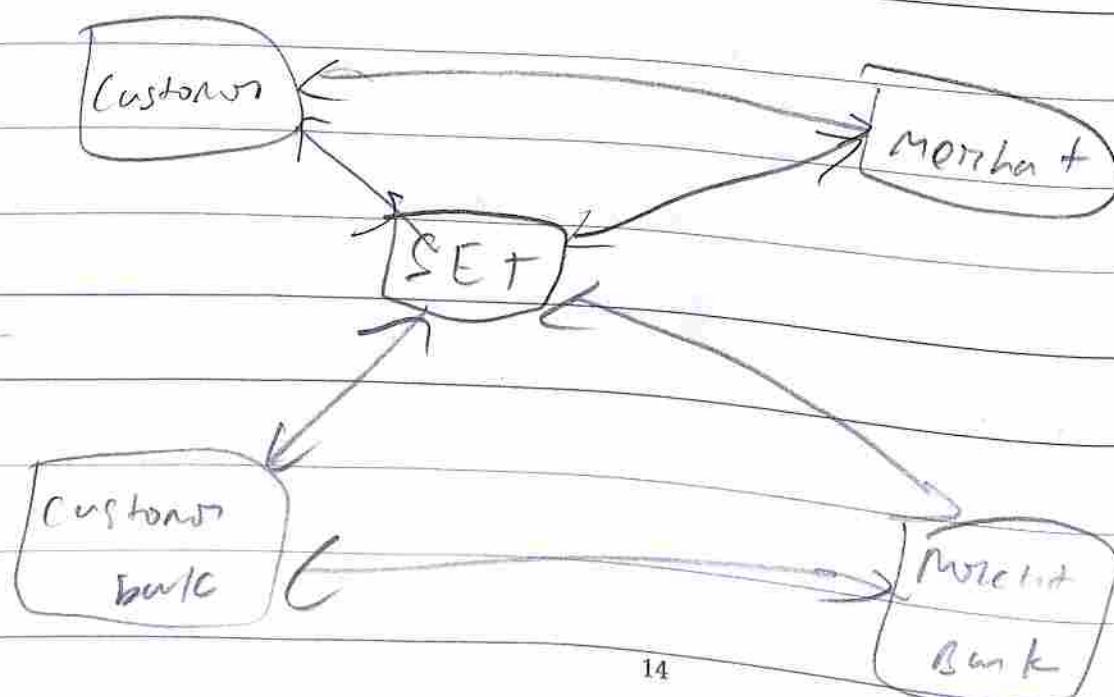
## B) SET

Secure Electronic Transaction.

~~Illustration~~ SET is the payment gateway between ~~digital commerce~~ 2 users.

- We use SET to ensure every transaction is completely secure and not tampered with.

- We keep SET as the communication of payment between 2 people so that transaction is secured and not tampered with.



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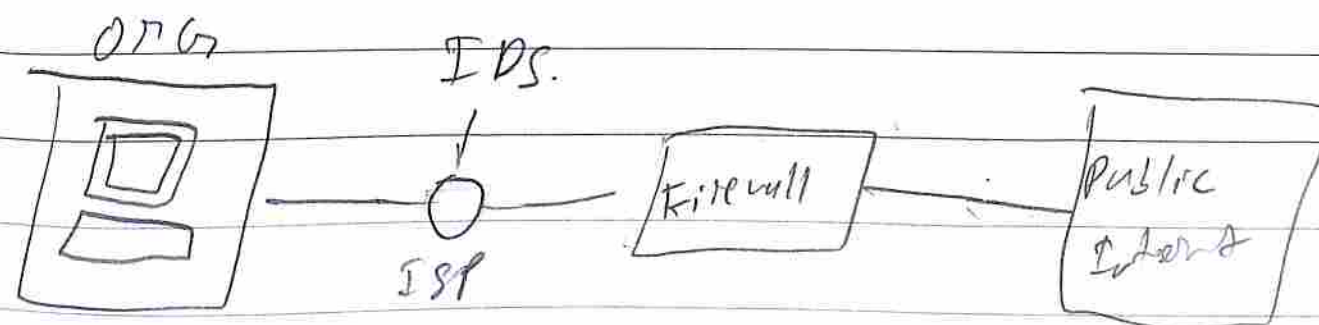
## B) IPS

Intrusion Detection System.

An IPS is used to filter in the fire wall for any malicious activity, if any is found, it will be caught by IPS.

If any activity happens and IPS is triggered, it will alert the concerned individual so they can deal with it.

IPS is a very good filtering system as the firewall as some times attackers will try various methods to gain access. But all of this activity is logged in the IPS.



## AI IP Sec

• IP Sec is a network layer security.

• It secures the IP address as well as the data.

• It encrypts the private IP data and sends it.

• It helps in tunneling the data connection so that it is only access by ~~sender~~ sender and receiver.

• It ensures data security, Confidentiality, integrity in the network layer. It works as its network layer security blanket.

• IP Sec plays an important role for security and encrypts it.





